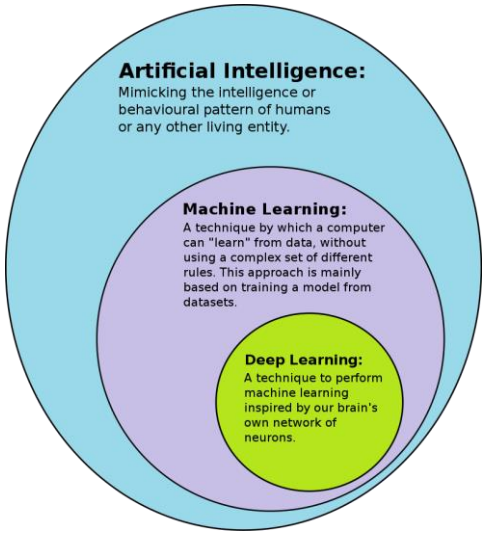
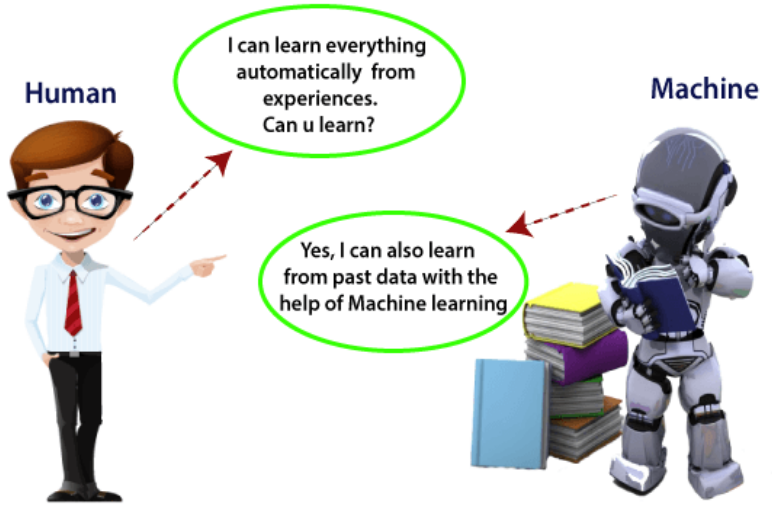
CS 736
Machine Learning (3-1-0)

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Dept. of Computer Science and Engg.
National Institute of Technology Karnataka
Surathkal, Mangalore

What is Machine Learning ?

- Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959).
- Concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own.
- One of the most popular AI techniques.
- Learning is all about finding useful abstractions or concepts that describe the world.

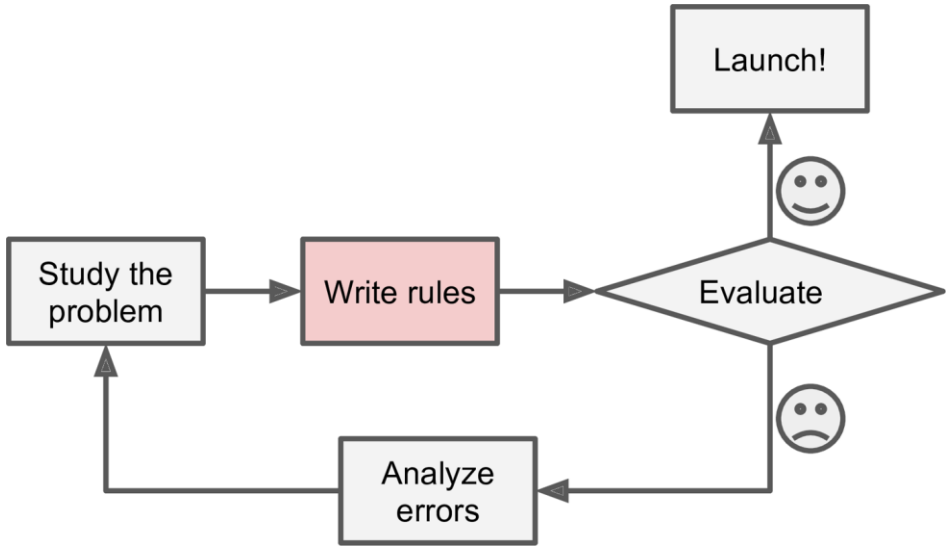




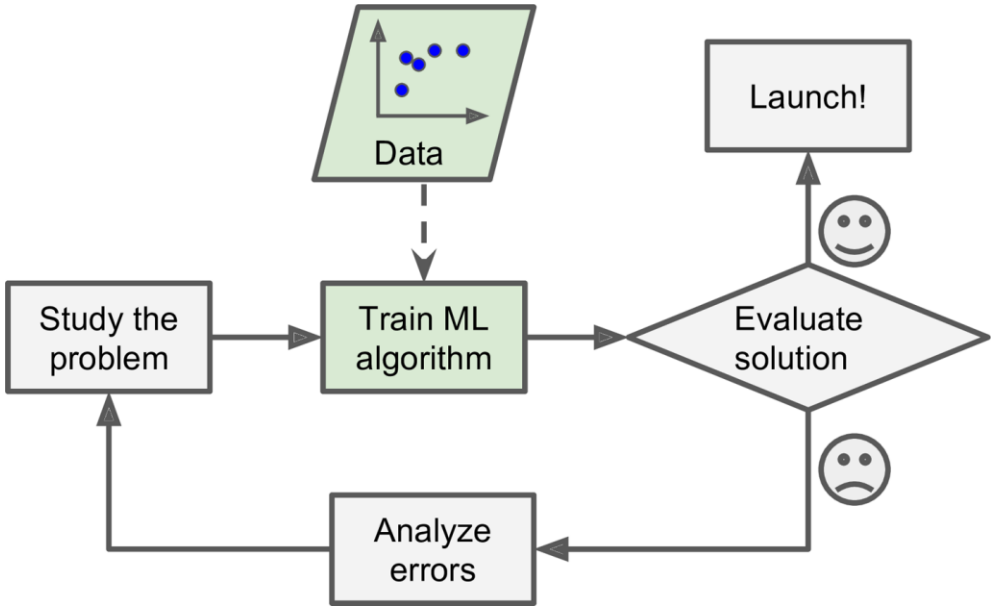
https://www.javatpoint.com/machine-learning

Machine learning methods learns from data.

- Learning from data is used in situations where we don't have an analytic solution, but we do have data that we can use to construct an empirical solution.
- Learning from data is one of the most widely used techniques in science, engineering, and economics, among other fields.
- Machine learning is the science and art of programming computers so they can learn from data.

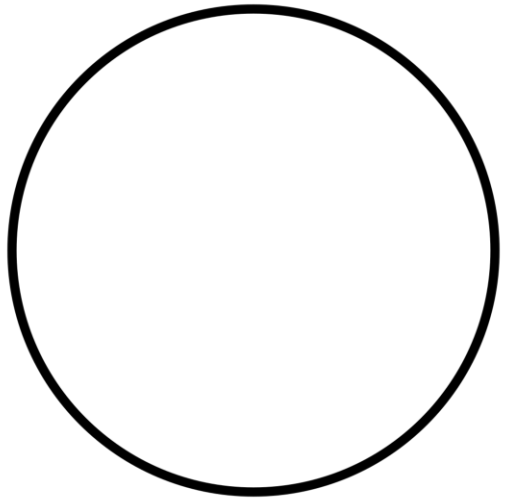


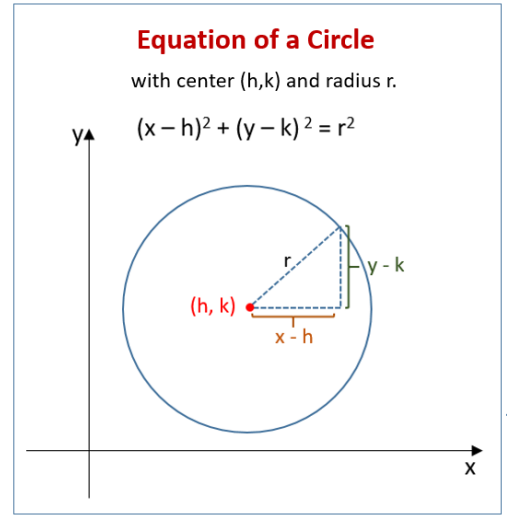
Traditional Approach



Machine Learning Approach

https://www.oreilly.com/library/view/hands-on-machine-learning/9781491962282/ch01.html





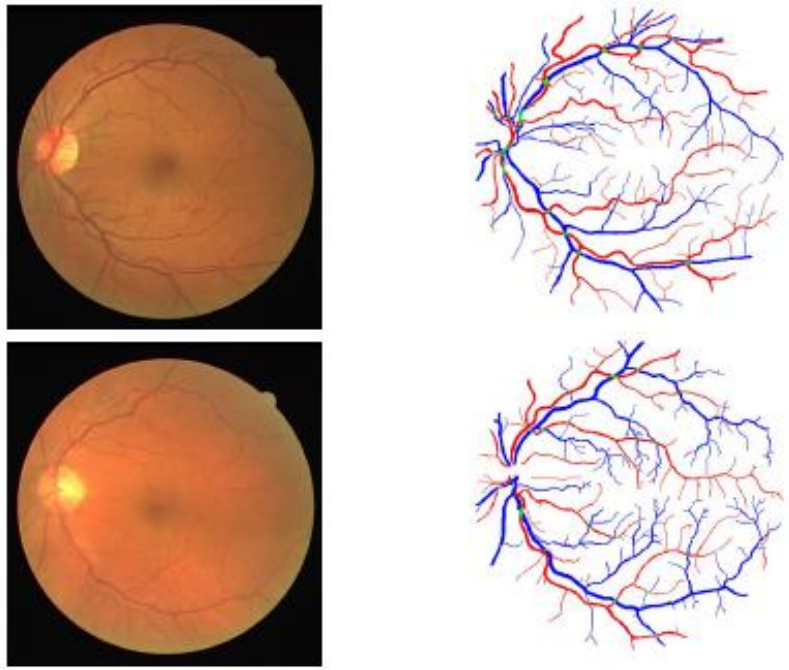


Equation of a Tree ?

We learned trees by looking at trees, not by studying its mathematical definition. In other words, we learned from data.

Why Machine Learning ?

- Machine learning is capable of doing tasks that are too complex for a person to implement directly.
- Useful to find solutions for complex problems for which there is no good solutions at all using a traditional approach.
- The importance of machine learning can be easily understood by its applications.
- Currently, machine learning is used in search engines, self-driving cars, cyber fraud detection, face recognition, spam filtering, medical image analysis etc.



automatic separation of arteries/veins

Course Syllabus

Introduction to Machine Learning, Different types of Learning, Evaluation metrices and cross validation, Overfitting and underfitting, Supervised learning, Linear regression, Logistic regression, Decision Trees, Support Vector Machines, Artificial neural networks, Unsupervised learning, Semi-supervised learning, Introduction to deep learning, Discussions and implementation of popular DL methods, Applications (image classification and segmentation).

Course Evaluation



Mid Sem – 20 %

End Sem – 40 %

Viva-voce – 20%

Assignments / Course Project/ New ideas – 20%

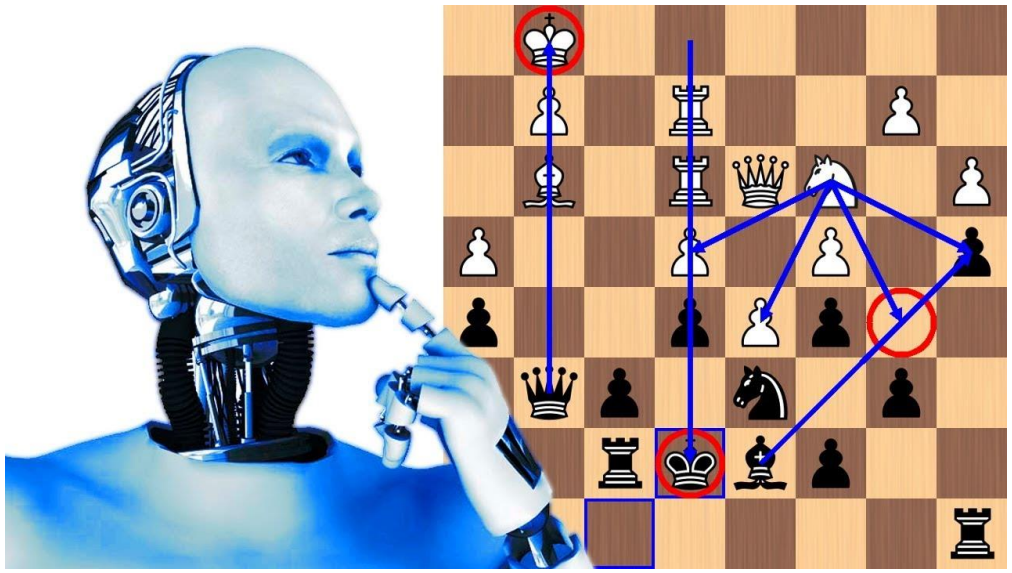
AI and machine learning

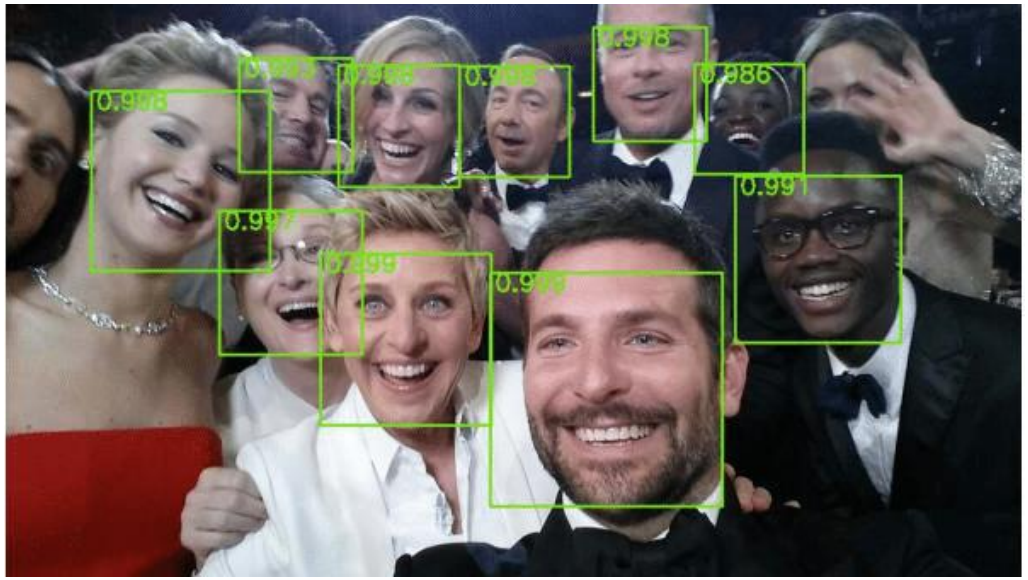
AI was introduced as an academic discipline in 1950s (Dartmouth Conference) to create machines that can think.

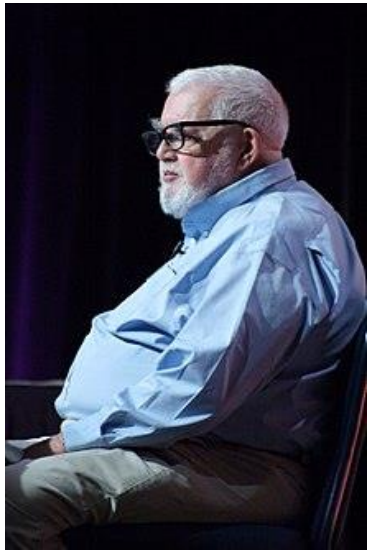
John McCarthy, widely recognized as one of the godfathers of AI, defined it as the science and engineering of making intelligent machines.

After Dartmouth, AI research was dominated by symbolic or rule-based approaches. However, it required manual rule creation.

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959).



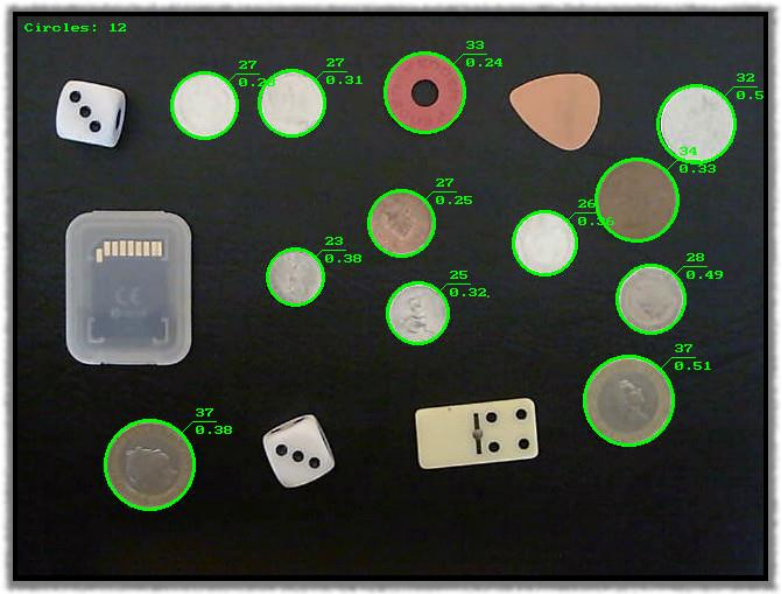


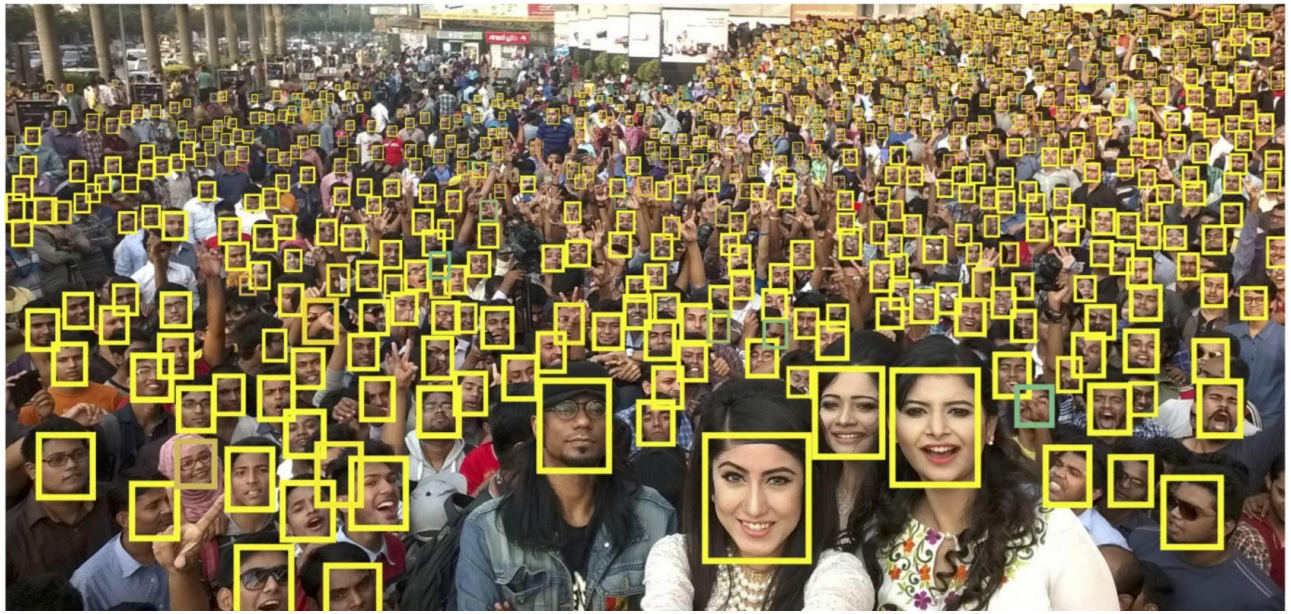


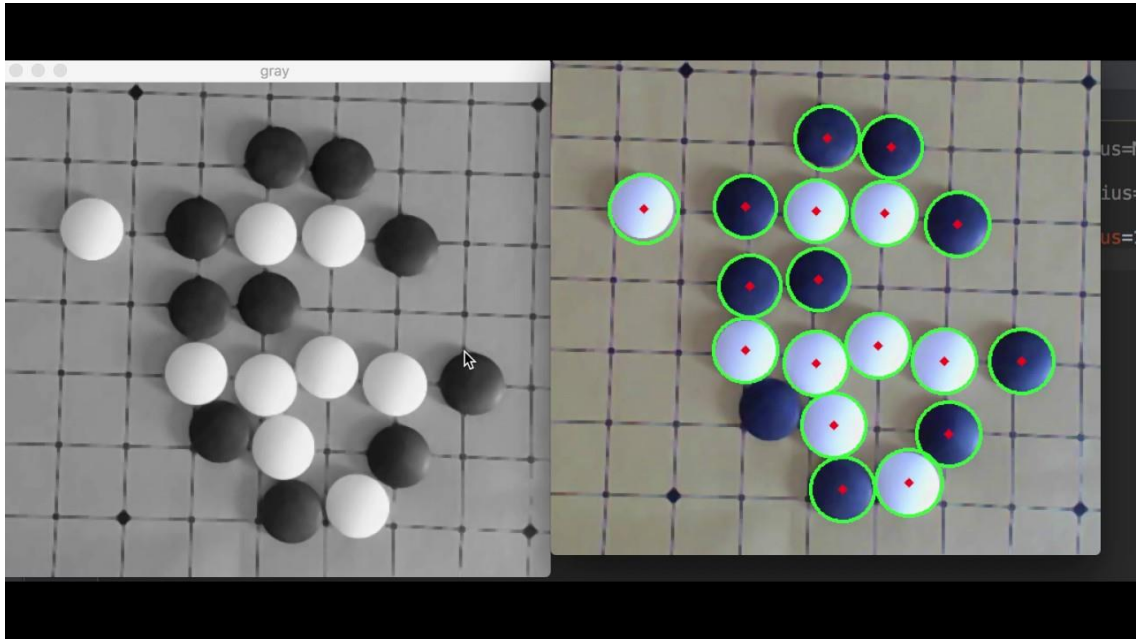
John McCarthy

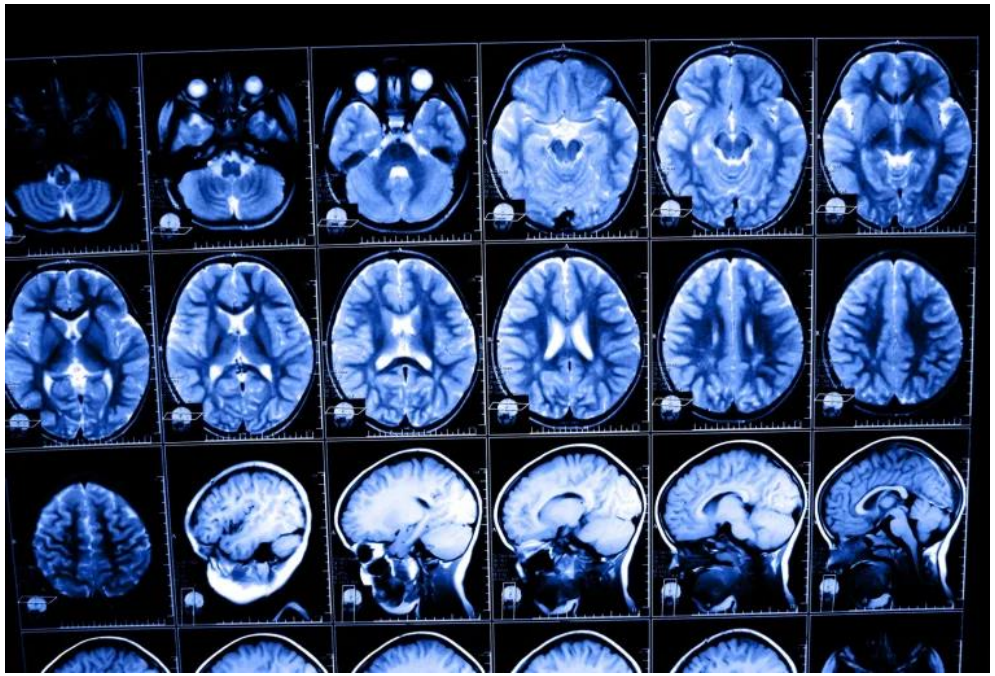


Arthur Lee Samuel


Circles: 12
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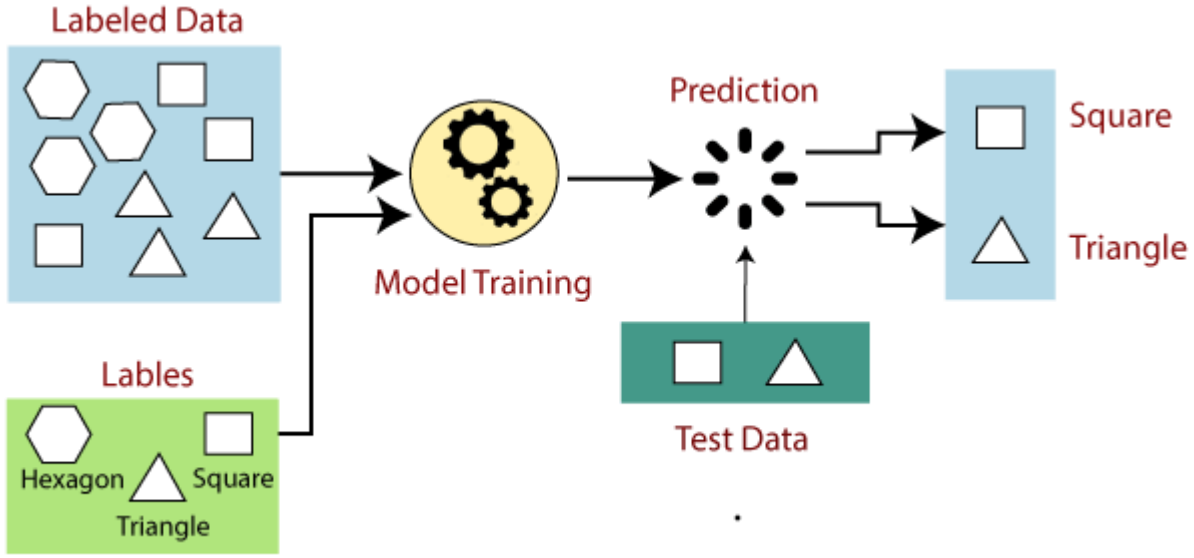



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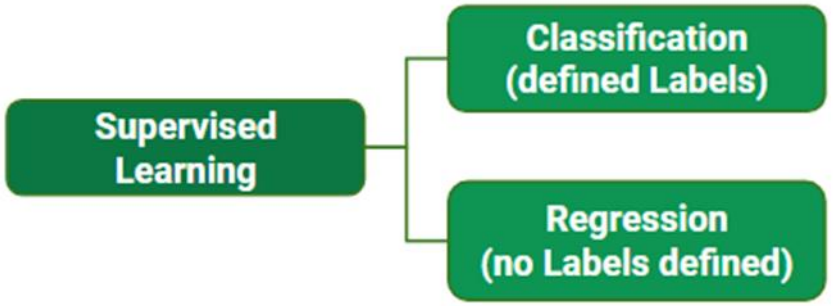
Types of Learning

- **Supervised learning** is the machine learning task of learning a function that maps an input to an output based on example input-output pairs.
- It infers a function from *labeled* *training data* consisting of a set of *training examples*.
- In supervised learning, each example is a *pair* consisting of an input object (typically a vector) and a desired output value (also called the *supervisory signal*).



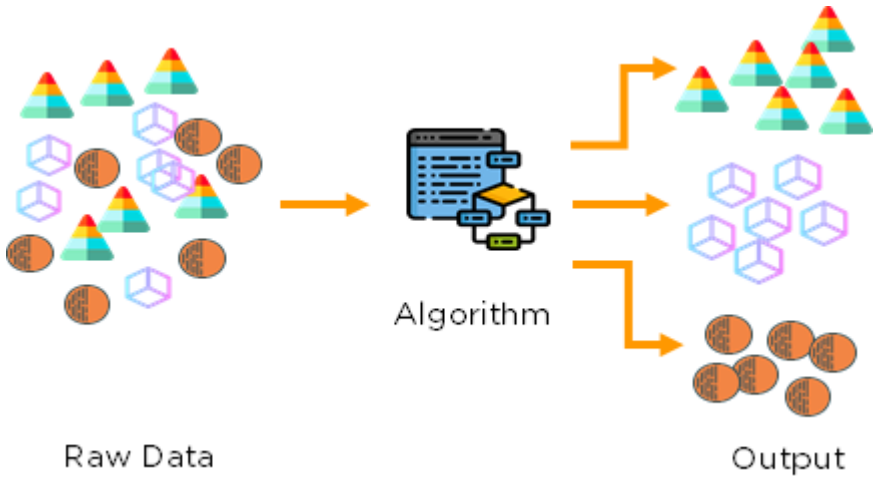
Some important supervised learning methods

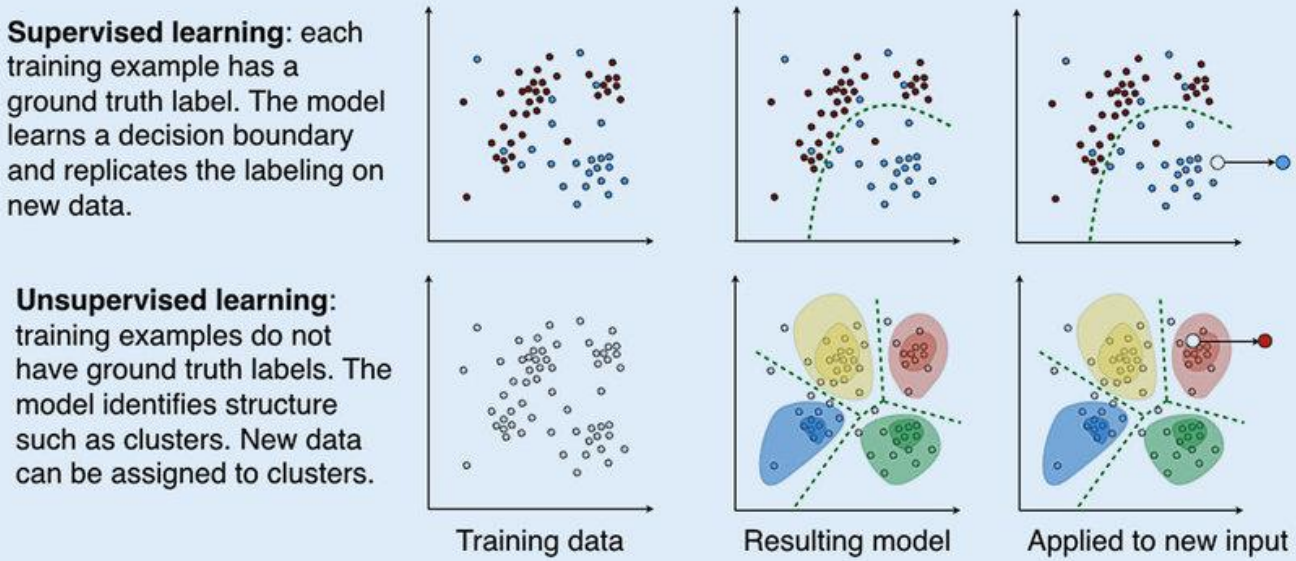
- Linear regression
- Logistic regression
- Support Vector Machines (SVM)
- Decision Trees and Random Forests
- Neural Networks



Unsupervised Learning

- In the unsupervised setting, the training data does not contain any output information at all.
- In contrast to supervised learning, it's not always easy to come up with metrics for how well an unsupervised learning algorithm is doing.





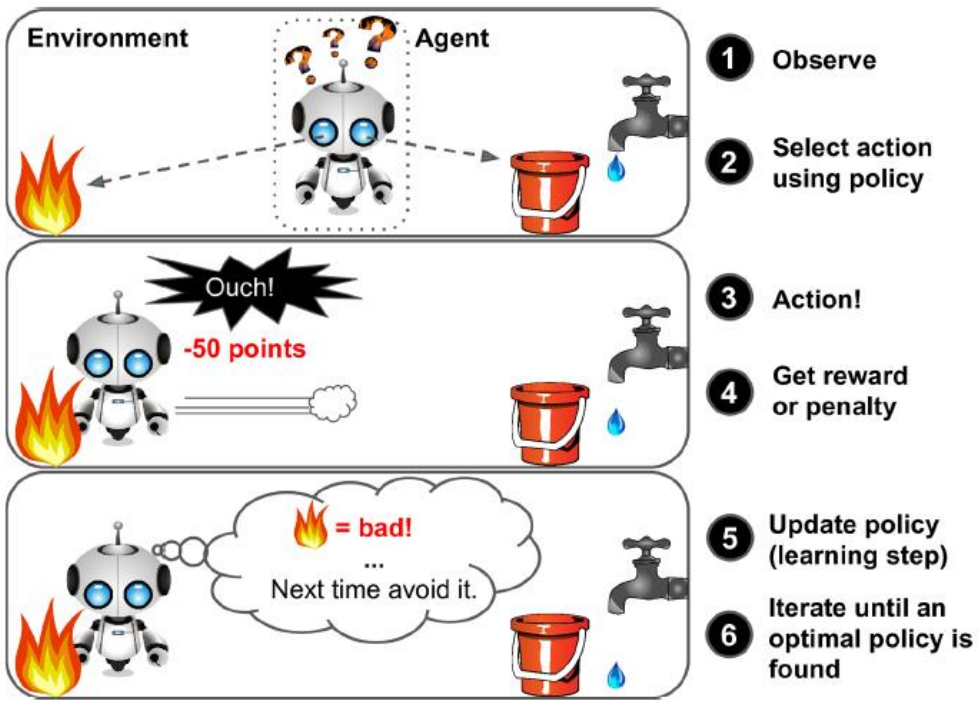
Some important unsupervised learning methods

- Clustering
- Dimensionality reduction

Reinforcement Learning

In reinforcement learning, the learning system, called an agent observe the environment, select and perform actions, and get rewards in return (or penalties in the form of negative rewards).

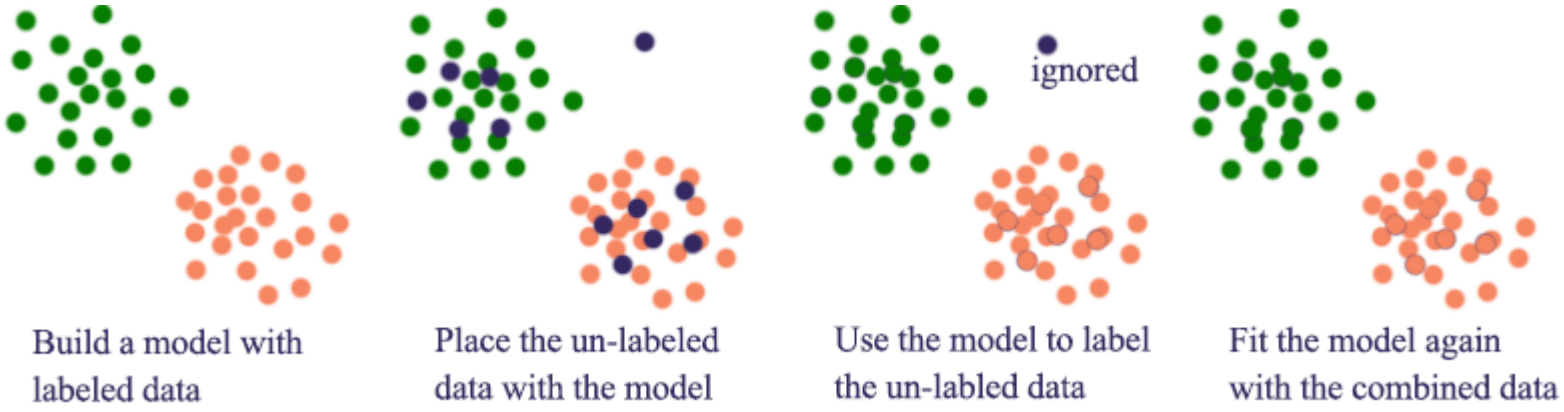
It must then learn by itself what is the best strategy, called a policy, to get the most reward over time. A policy defines what action the agent should choose when it is in a given situation.



Semi supervised Learning

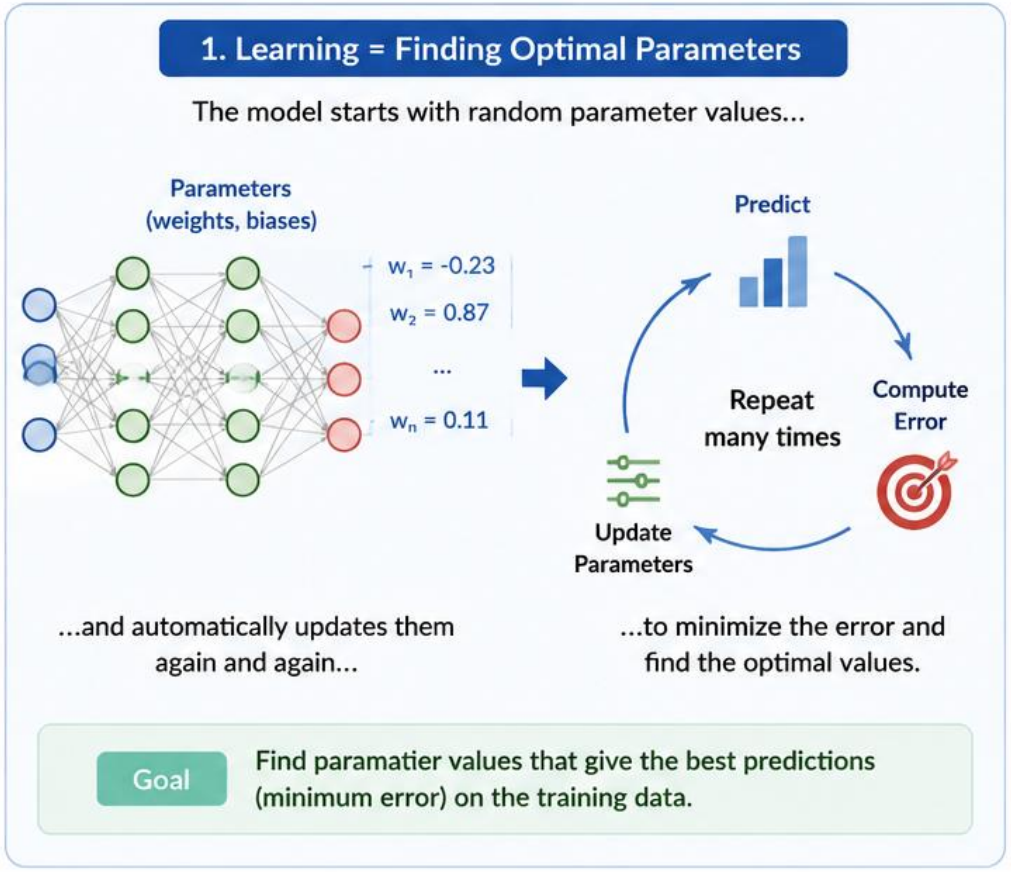
Some algorithms can deal with partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data. This is called *semi-supervised learning.*

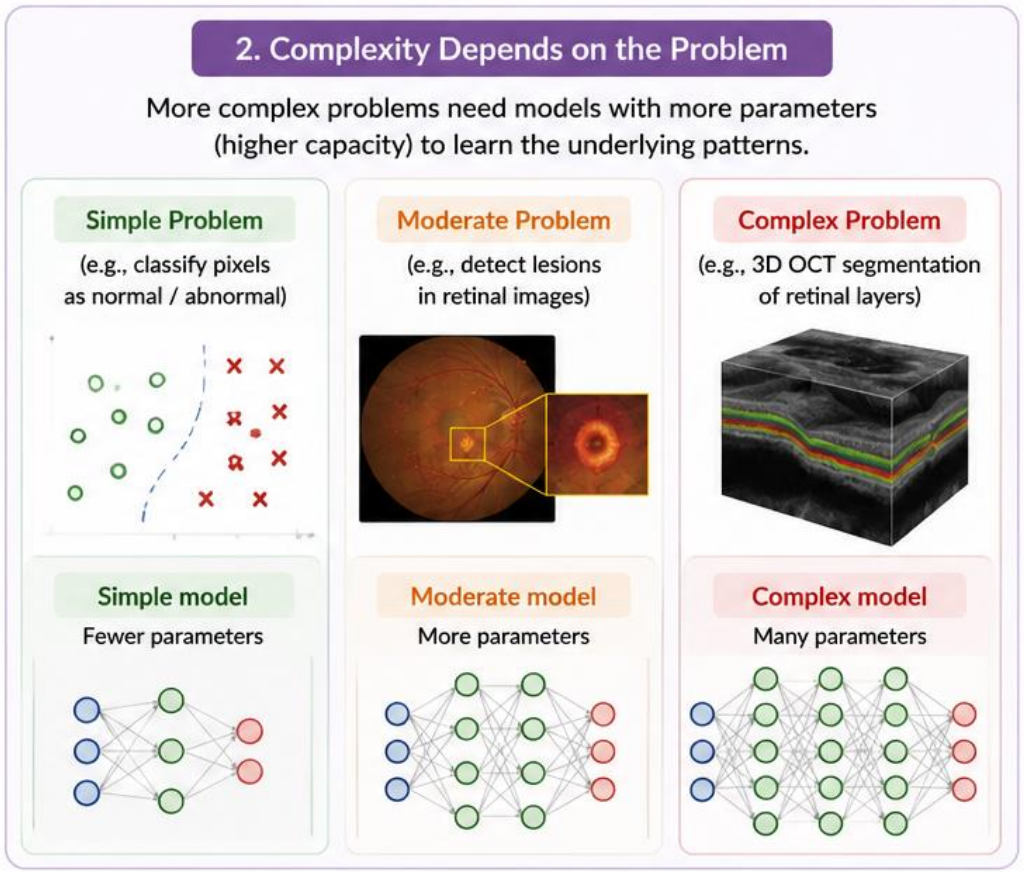
Most semi-supervised learning algorithms are combinations of unsupervised and supervised algorithms.



What happens during training?

- Learning means automatically finding the optimal parameter values that minimize prediction error.
- More complex problems generally require models with greater capacity—and hence more parameters—to learn the underlying patterns.





Main challenges of machine learning

Insufficient quantity of training data

- Many machine learning algorithms require large amounts of data before they begin to give useful results.
- Even for very simple problems you typically need thousands of examples, and for complex problems such as image or speech recognition you may need millions of examples (unless you can reuse parts of an existing model).

The Unreasonable Effectiveness of Data

In a famous paper published in 2001, Microsoft researchers Michele Banko and Eric Brill showed that very different Machine Learning algorithms, including fairly simple ones, performed almost identically well on a complex problem of natural language disambiguation[8] once they were given enough data (as you can see in Figure 1-20).

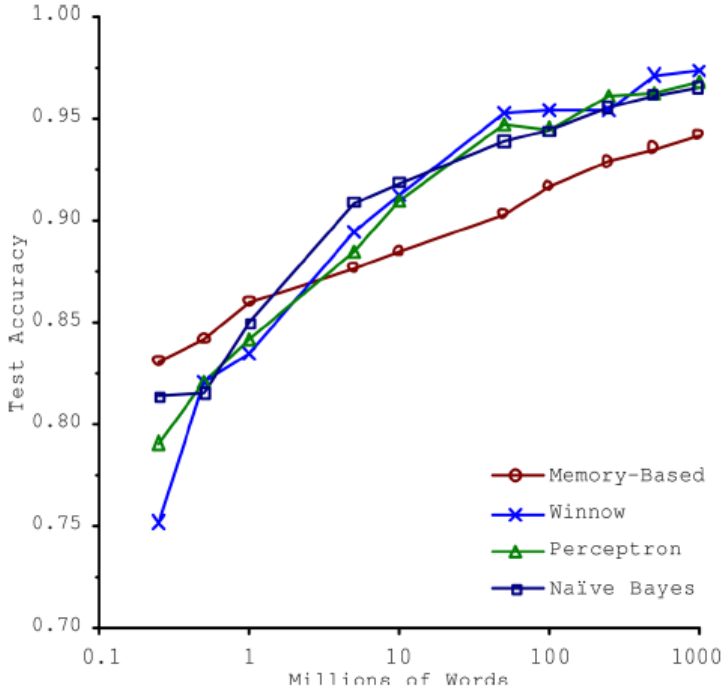
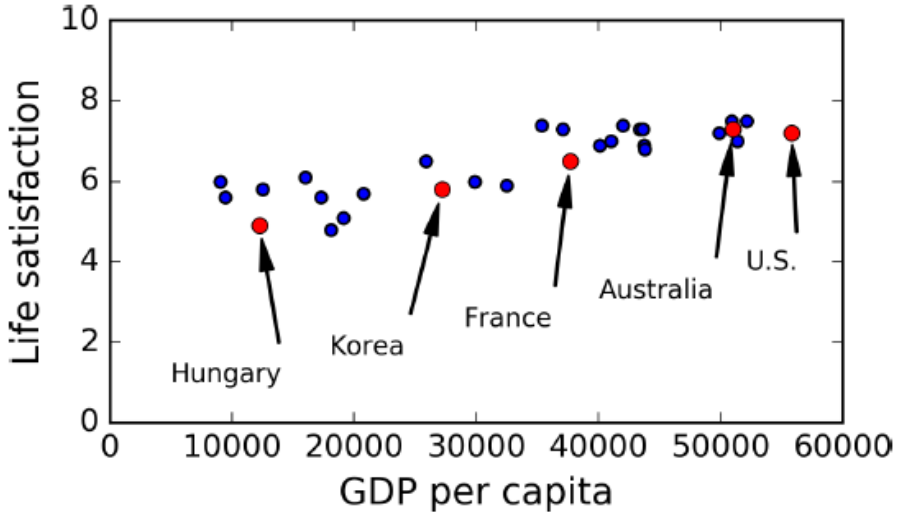
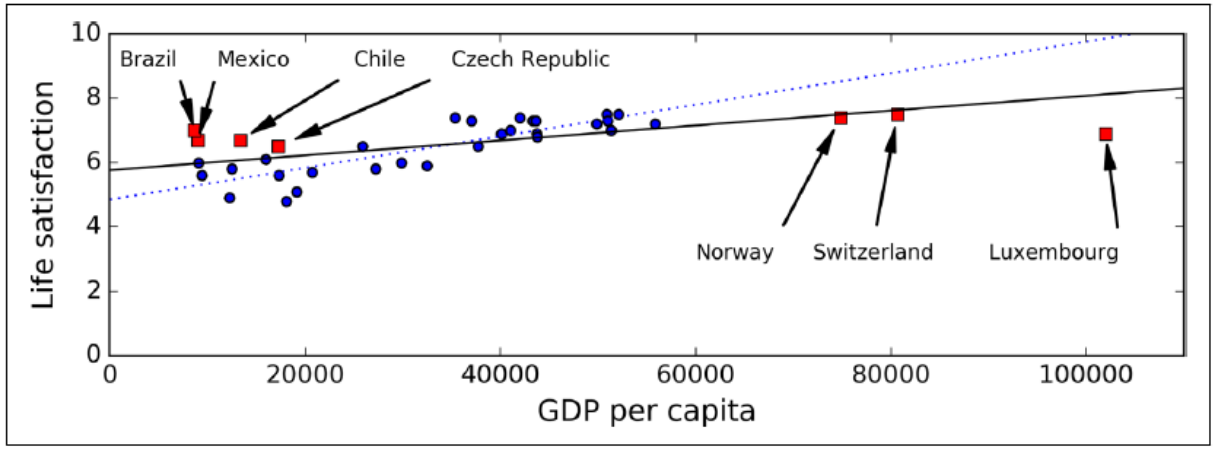


Figure 1-20. The importance of data versus algorithms[9]

Nonrepresentative Training data

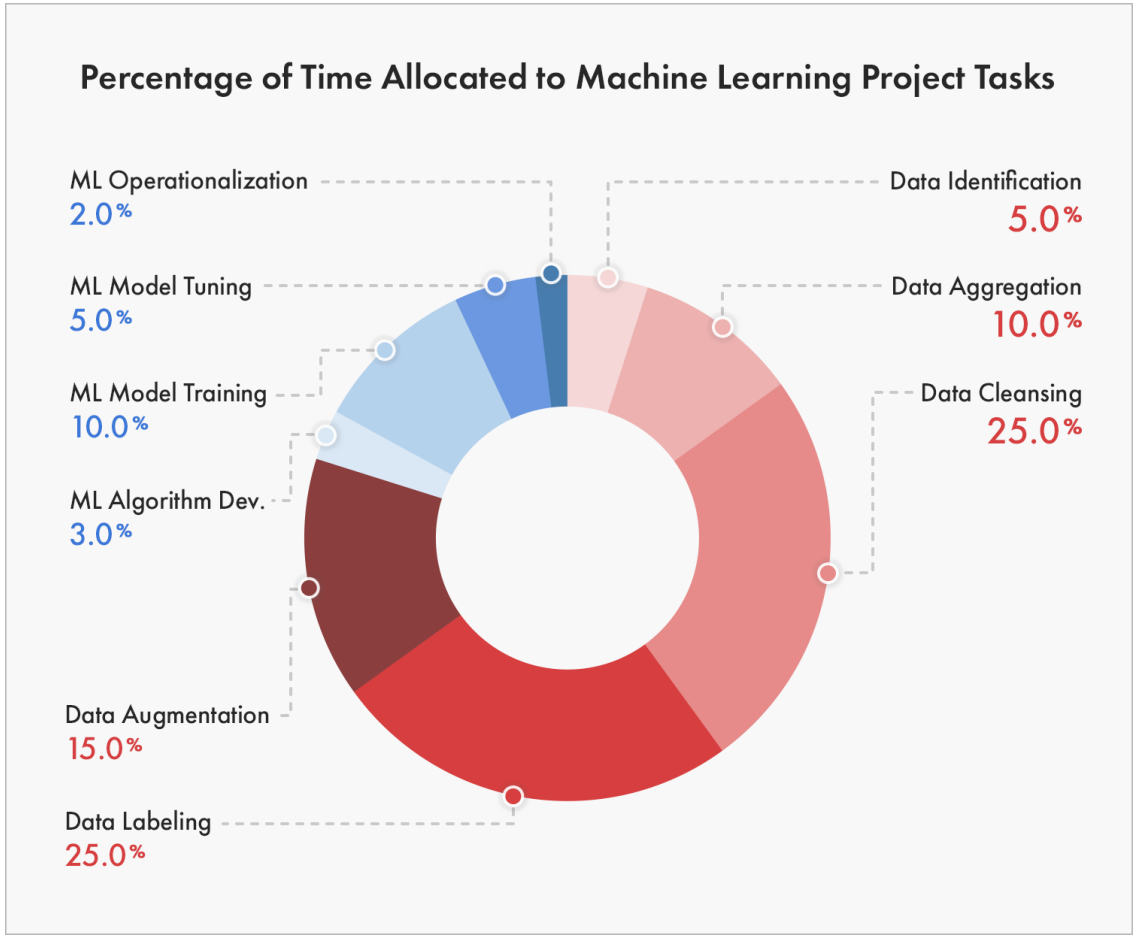
- In order to generalize well, it is crucial that your training data be representative of the new cases you want to generalize to.
- It is crucial to use a training set that is representative of the cases you want to generalize to.
- If the sample is too small, you will have *sampling noise* (i.e., nonrepresentative data as a result of chance), but even very large samples can be nonrepresentative if the sampling method is flawed. This is called *sampling bias*.





Poor-Quality data

- If your training data is full of errors, outliers, and noise (e.g., due to poor quality measurements), it will make it harder for the system to detect the underlying patterns, so your system is less likely to perform well.
- Clean the data before using it for training/testing
- If some instances are clearly outliers, it may help to simply discard them or try to fix the errors manually.

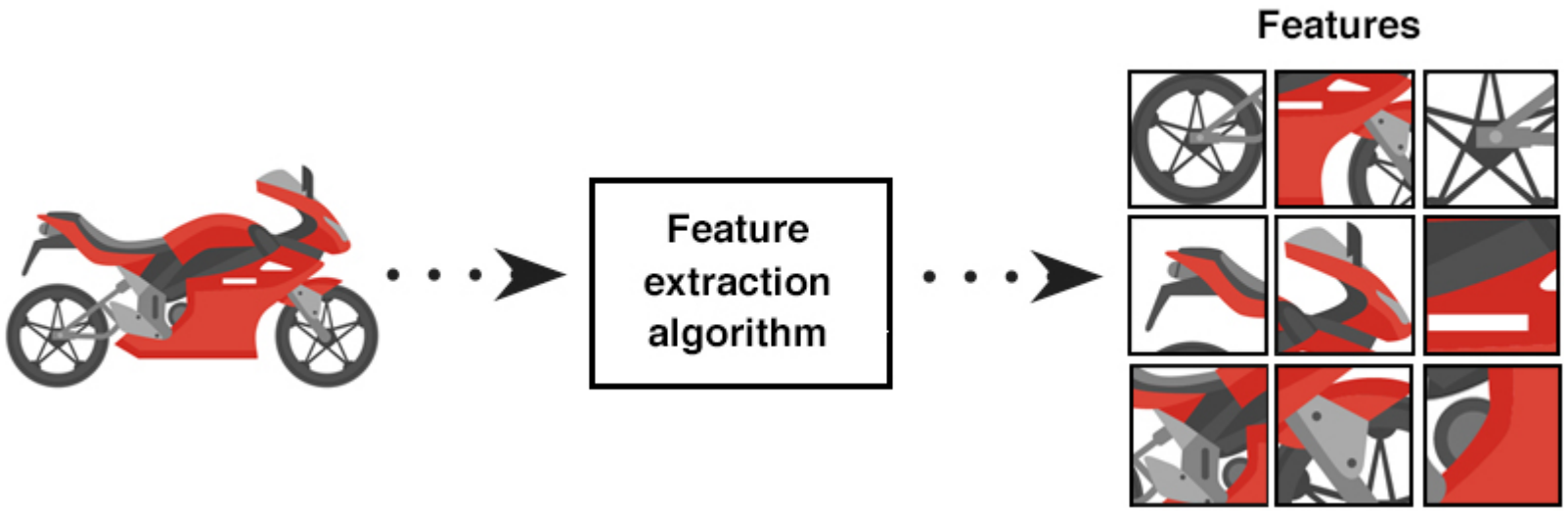


Irrelevant Features

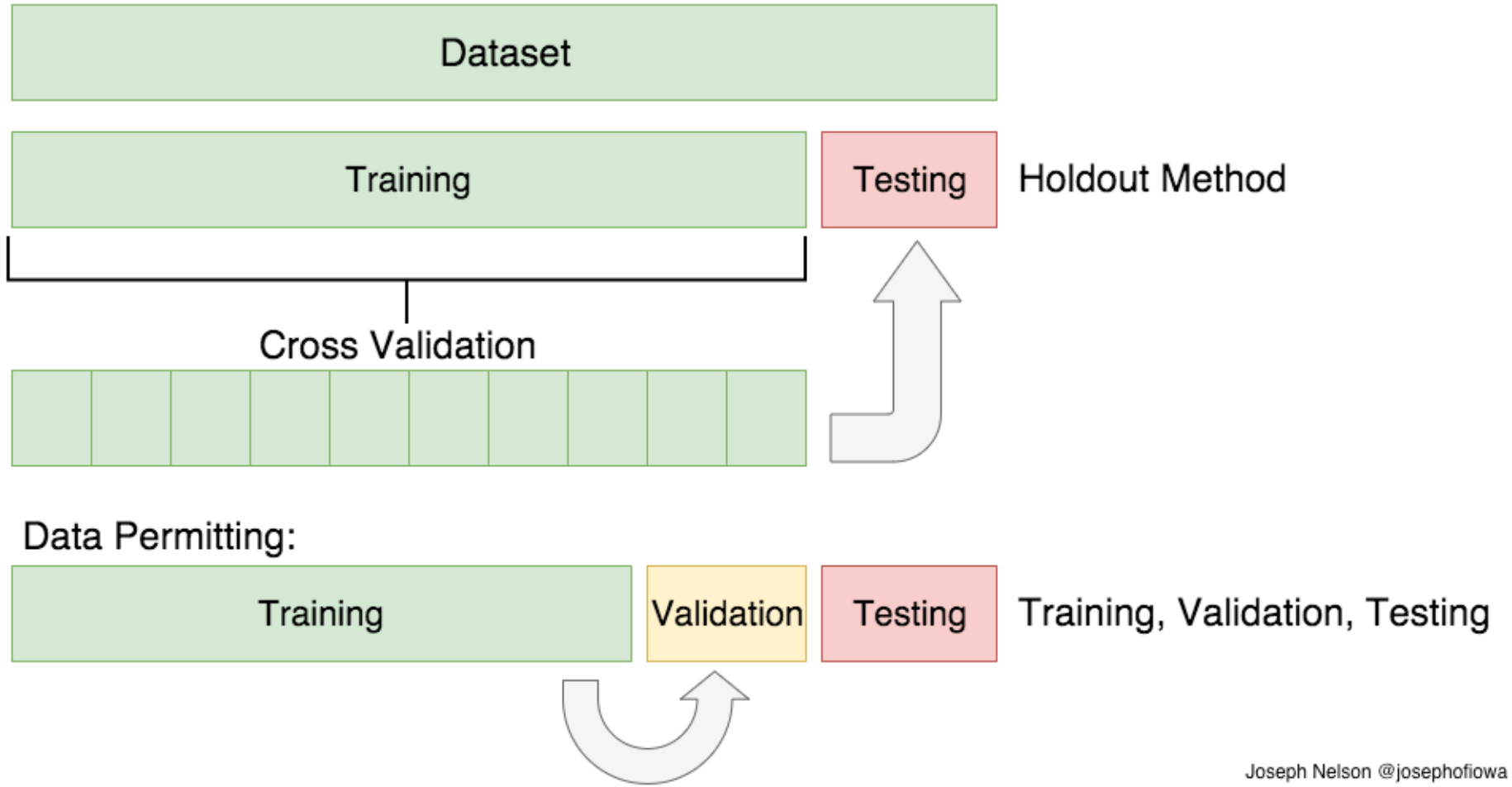
A critical part of the success of a Machine Learning project is coming up with a good set of features to train on.

This process, called *feature engineering*, involves:

- *Feature selection*: selecting the most useful features to train on among existing features.
- *Feature extraction*: combining existing features to produce a more useful one (as we saw earlier, dimensionality reduction algorithms can help).
- Creating new features by gathering new data



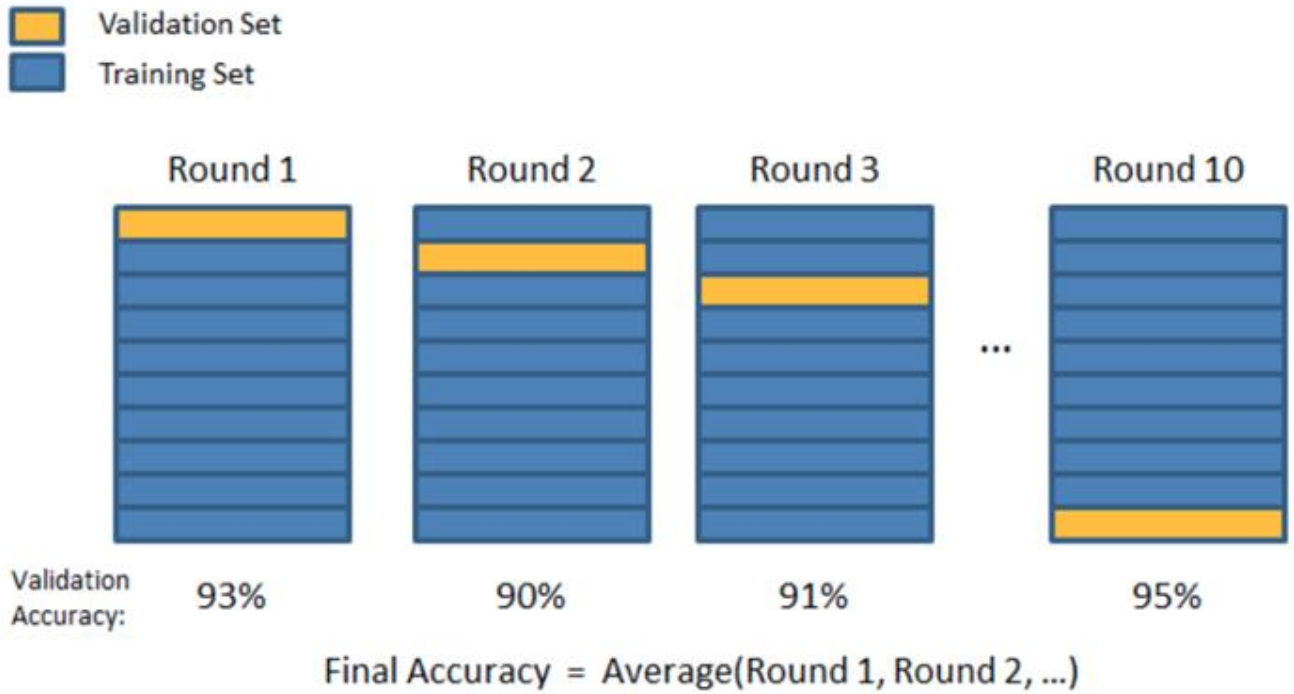
Training, Validation and Test data



https://towardsdatascience.com/train-test-split-and-cross-validation-in-python-80b61beca4b6

K-fold cross validation

- In K-Folds Cross Validation we split our data into k different subsets (or folds).
- We use k-1 subsets to train our data and leave the last subset (or the last fold) as test data.
- We then average the model against each of the folds and then finalize our model. After that we test it against the test set.



https://towardsdatascience.com/train-test-split-and-cross-validation-in-python-80b61beca4b6

Overfitting and underfitting

Overfitting : The model performs well on the training data, but does not generalize well.

Underfitting : The model is too simple to learn the underlying structure of the data.

